

Assignment 7 of MATP4820

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Problem 1

Consider the problem

$$\min_{x,y \in \mathbb{R}} f(x,y) = \frac{1}{2}x^2 - xy + 4y^2 + 5x - 12y, \text{ s.t. } x \leq 2, y \leq 4.$$

1. Does $(0, 1)$ satisfy the optimality condition?

First,

$$\nabla f(x, y) = \begin{bmatrix} x - y + 5 \\ -x + 8y - 12 \end{bmatrix}$$

$$\nabla f(0, 1) = \begin{bmatrix} 4 \\ -4 \end{bmatrix}$$

Now optimality condition:

$$\nabla f(\bar{x})^T (x - \bar{x}) = \begin{bmatrix} 4 \\ -4 \end{bmatrix}^T \left(\begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) \geq 0$$

$$\implies 4(x) - 4(y - 1) \geq 0$$

$$\implies 4x - 4y + 4 \geq 0$$

This condition does not hold for $x = 0, y = 3$

$$4x - 4y + 4 \geq 0 \implies 4(0) - 4(3) + 4 = -8 < 0.$$

Thus the optimality condition does not hold.

2. Show that f is convex on $\mathbb{R}^2$. Find the global minimizer.

First we have

$$\nabla f(x, y) = \begin{bmatrix} x - y + 5 \\ -x + 8y - 12 \end{bmatrix}$$

$$\nabla^2 f(x, y) = \begin{bmatrix} 1 & -1 \\ -1 & 8 \end{bmatrix}$$

$$\det(\nabla^2 f(x, y)) = (1)(8) - (-1)(-1) = 8 - 1 = 7 > 0$$

Thus, $\nabla^2 f(x, y)$ is symmetric, and its determinant is positive, so we can see it is positive definite, and f is convex on $\mathbb{R}^2$.

Now,

$$\nabla f(x, y) = \begin{bmatrix} x - y + 5 \\ -x + 8y - 12 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\implies 7y - 7 = 0 \implies y = 1 \implies x = -4$$

$$x = -4 < 2, y = 1 < 4 \implies \text{Feasible Point}$$

Therefore, we have local minimizer of $x = -4, y = 1$ (equivalently $\begin{bmatrix} -4 \\ 1 \end{bmatrix}$ or $(-4, 1)$).

Since we showed f is convex on $\mathbb{R}^2$, this point is the global minimizer.

Problem 2

Consider the problem

$$\begin{aligned} \min_{x,y \in \mathbb{R}} f(x,y) &= -x^2 + xy + \frac{1}{2}y^2 - 2x - y \\ \text{s.t. } x &\geq -1, y \geq -1, x + y \leq 4. \end{aligned}$$

Find all KKT points and the global minimizer.

Written in standard form

$$\min_{x,y \in \mathbb{R}} f(x,y) = -x^2 + xy + \frac{1}{2}y^2 - 2x - y$$

$$\text{s.t. } g_1(x,y) = -x_1 - 1 \leq 0$$

$$g_2(x,y) = -y - 1 \leq 0$$

$$g_3(x,y) = x + y - 4 \leq 0$$

$$\nabla f(x,y) = \begin{bmatrix} -2x + y - 2 \\ x + y - 1 \end{bmatrix}$$

$$\nabla g_1 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \nabla g_2 = \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \nabla g_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

For KKT points,

$$\nabla f(\bar{x}, \bar{y}) + \sum_{i=1}^3 u_i \nabla g_i(\bar{x}, \bar{y}) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$u_i \geq 0, \quad \forall i = 1, 2, 3$$

$$g_i(\bar{x}, \bar{y}) \leq 0, \quad \forall i = 1, 2, 3$$

$$u_i g_i(\bar{x}, \bar{y}) = 0, \quad \forall i = 1, 2, 3$$

Now,

$$\begin{aligned} \begin{bmatrix} -2x + y - 2 \\ x + y - 1 \end{bmatrix} + u_1 \begin{bmatrix} -1 \\ 0 \end{bmatrix} + u_2 \begin{bmatrix} 0 \\ -1 \end{bmatrix} + u_3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \implies \begin{bmatrix} -2x + y - 2 - u_1 + u_3 \\ x + y - 1 - u_2 + u_3 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ u_i &\geq 0, \quad i = 1, 2, 3 \end{aligned}$$

$$-x - 1 \leq 0, \quad -y - 1 \leq 0, \quad x + y - 4 \leq 0$$

$$u_1(-x - 1) = 0, \quad u_2(-y - 1) = 0, \quad u_3(x + y - 4) = 0$$

Case 1 interior point: $(u_i = 0, \forall i = 1, 2, 3)$

$$\begin{aligned} \nabla f(\bar{x}, \bar{y}) &= \begin{bmatrix} -2x + y - 2 \\ x + y - 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \implies -3x - 1 &= 0 \implies x = -\frac{1}{3}, y = \frac{4}{3} \end{aligned}$$

$$x = -\frac{1}{3} \geq -1, \quad y = \frac{4}{3}, \quad x + y = 1 \leq 4 \implies (\text{Feasible Point})$$

Since $u_1, u_2, u_3 = 0$ and the point $(-\frac{1}{3}, \frac{4}{3})$ is feasible, we have a KKT point: $(-\frac{1}{3}, \frac{4}{3})$

Case 2 $(x + y - 4 = 0)$ active: $(u_1, u_2 = 0, u_3 \geq 0)$

$$\nabla f(\bar{x}, \bar{y}) = \begin{bmatrix} -2x + y - 2 + u_3 \\ x + y - 1 + u_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

with $y = 4 - x$

$$x + y - 1 + u_3 = 0 \implies x + 4 - x + 1 + u_3 = 0 \implies 3 + u_3 = 0 \implies u_3 = -3 < 0$$

But $u_3 \geq 0$, thus there can't be a KKT point in this case.

Case 3 ($-x - 1 = 0$) active: ($u_2, u_3 = 0, u_1 \geq 0$)

$$\nabla f(\bar{x}, \bar{y}) = \begin{bmatrix} -2x + y - 2 - u_1 \\ x + y - 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

with $x = -1$

$$\begin{bmatrix} -2(-1) + y - 2 - u_1 \\ -1 + y - 1 \end{bmatrix} = \begin{bmatrix} y - u_1 \\ y - 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\implies y = 2 \implies 2 - u_1 = 0 \implies u_1 = 2 \geq 0$$

Now,

$$x = -1 \geq -1, \quad y = 2 \geq -1, \quad x + y = 1 \leq 4 \implies (\text{Feasible Point})$$

Since $u_1 \geq 0$ and the point $(-1, 2)$ is feasible, we have a KKT point $(-1, 2)$

Case 4 ($-y - 1 = 0$) active: ($u_2, u_3 = 0, u_1 \geq 0$)

$$\nabla f(\bar{x}, \bar{y}) = \begin{bmatrix} -2x + y - 2 \\ x + y - 1 - u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

with $y = -1$

$$\begin{bmatrix} -2x - 1 - 2 \\ x - 1 - 1 - u_2 \end{bmatrix} = \begin{bmatrix} -2x - 3 \\ x - u_2 - 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\implies 2u_2 - 7 = 0 \implies u_2 = -\frac{7}{2} < 0$$

But $u_2 \geq 0$, thus there can't be a KKT point in this case.

Case 5 boundary ($-x - 1 = 0, -y - 1 = 0$): ($u_1, u_2 \geq 0, u_3 = 0$)

With $x = -1, y = -1$

$$\nabla f(-1, -1) = \begin{bmatrix} -2(-1) - 1 - 2 \\ -1 - 1 - 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -3 \end{bmatrix}$$

Now,

$$\begin{bmatrix} -1 \\ -3 \end{bmatrix} + u_1 \begin{bmatrix} -1 \\ 0 \end{bmatrix} + u_2 \begin{bmatrix} 0 \\ -1 \end{bmatrix} + u_3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\implies \begin{bmatrix} -1 \\ -3 \end{bmatrix} + u_1 \begin{bmatrix} -1 \\ 0 \end{bmatrix} + u_2 \begin{bmatrix} 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

But $u_1, u_2 \geq 0$, thus there can't be a KKT point in this case.

Case 6 boundary ($x + y - 4 = 0, y + 1 = 0$) active: ($u_1 = 0, u_2, u_3 \geq 0$)

With $x + y = 4, y + 1 = 0 \implies x = 5, y = -1$.

$$\nabla f(5, -1) = \begin{bmatrix} -2(5) + (-1) - 2 \\ 5 + (-1) - 1 \end{bmatrix} = \begin{bmatrix} -13 \\ 3 \end{bmatrix}$$

Now,

$$\begin{bmatrix} -13 \\ 3 \end{bmatrix} + u_2 \begin{bmatrix} 0 \\ -1 \end{bmatrix} + u_3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\implies -13 + u_3 = 0 \implies u_3 = 13 \geq 0$$

$$\implies 3 - u_2 + u_3 = 0 \implies 3 - u_2 + 13 = 0 \implies u_2 = 16 \geq 0$$

Since $u_2, u_3 \geq 0$ and the point $(5, -1)$ is feasible, we have a KKT point: $(5, -1)$

Case 7 Boundary ($x + y - 4 = 0, x + 1 = 0$) active: ($u_2 = 0, u_1, u_3 \geq 0$)

With $x + y = 4, x = -1 \implies x = -1, y = 5$

$$\nabla f(-1, 5) = \begin{bmatrix} -2(-1) + 5 - 2 \\ -1 + 5 - 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

Now,

$$\begin{bmatrix} 5 \\ 3 \end{bmatrix} + u_1 \begin{bmatrix} -1 \\ 0 \end{bmatrix} + u_3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\implies 3 + u_3 = 0 \implies u_3 = -3 < 0$$

But $u_3 \geq 0$, thus there can't be a KKT point in this case.

Final Set of KKT points: $(-\frac{1}{3}, \frac{4}{3}), (-1, 2), (5, -1)$

Now finding global min:

We have KKT points $(-\frac{1}{3}, \frac{4}{3}), (-1, 2), (5, -1)$

Evaluating f at these points:

$$f(-\frac{1}{3}, \frac{4}{3}) = -(-\frac{1}{3})^2 - \frac{1}{3}(\frac{4}{3}) + \frac{1}{2}(\frac{4}{3})^2 - 2(-\frac{1}{3}) - (\frac{4}{3}) = -\frac{1}{3}$$

$$f(-1, 2) = -(-1)^2 - 1(2) + \frac{1}{2}(2)^2 - 2(-1) - 2 = -1$$

$$f(5, -1) = -(5)^2 + 5(-1) + \frac{1}{2}(-1)^2 - 2(5) - (-1) = -38.5$$

Thus, $f(5, -1) < f(-1, 2) < f(-\frac{1}{3}, \frac{4}{3})$, so $(5, -1)$ is the global min

Problem 3

Consider the nonnegative quadratic program:

$$\underset{\mathbf{x} \in X}{\text{minimize}} f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} - \mathbf{c}^\top \mathbf{x}, \text{ s.t. } \mathbf{A} \mathbf{x} = \mathbf{b} \quad (1)$$

where $\mathbf{Q}$ is a symmetric and positive semidefinite matrix, and $X = \{\mathbf{x} \in \mathbb{R}^n : x_i \geq 0, i = 1, \dots, n\}$.

Use the instructor's provided file `penalty_qp.m` to write a Matlab function `penalty_qp` with input $\mathbf{Q}, \mathbf{c}, \mathbf{A}, \mathbf{b}$, initial vector $\mathbf{x}_0$, stopping tolerance `tol`, initial penalty parameter μ_0 , and the final μ_1 . Also test your function by running the provided test file `test_penalty_qp.m` and compare to the instructor's function. Print your code and the results you get. **[Do your best to optimize the efficiency of your code. You will lose 10% marks if your solver takes more than double of time of the instructor's solver, and lose 20% if your solver takes more than triple of the instructor's solver.]**

Quadratic Penalty Method Function

```
function [x, hist_obj, hist_res] = penalty_qp(Q,c,A,b,tol,mu0,mu1,x0)
% quadratic penalty method for the quadratic programming
% min_x 0.5*x'*Q*x - c'*x
% s.t.   x >= 0, A*x == b

mu = mu0;
x = x0;

% precompute AtA and Atb outside the loops
AtA = A'*A;
Atb = A'*b;

% compute the residual for the constraint A*x == b
r = A * x - b;

res = norm(r);
grad_err = 1;
hist_res = res;
hist_obj = 0.5*x'*Q*x - c'*x;
```

```

while (res > tol || grad_err > tol) && mu < mu1
    % use constant stepsize
    alpha = 1/norm(Q + mu*AtA);
    % compute the gradient
    grad = Q*x - c + mu*(AtA*x - Atb);

    % compute violation of optimality condition
    grad_err = norm(x - max(0, x - grad));
    while grad_err > tol
        % update x
        x = max(0, x - alpha*grad);

        % compute the gradient
        grad = Q*x - c + mu*(AtA*x - Atb);

        % compute violation of optimality condition
        grad_err = norm(x - max(0, x - grad));
    end
    % compute the residual
    r = A*x - b;
    res = norm(r);
    obj = 0.5*x'*Q*x - c'*x;

    % save res and obj
    hist_res = [hist_res; res];
    hist_obj = [hist_obj; obj];

    % increase the penalty parameter
    mu = 5*mu;
end
end

```

Code Results: `test_pg.m` Output

```
>> test_penalty_qp
```

Minimum found that satisfies the constraints.

Optimization completed because the objective function is non-decreasing in feasible directions, to within the value of the optimality tolerance, and constraints are satisfied to within the value of the constraint tolerance.

<stopping criteria details>

Student solver: Total running time of student code is 25.6695

Instructor solver: Total running time of instructor code is 34.7095

Student Figure Result

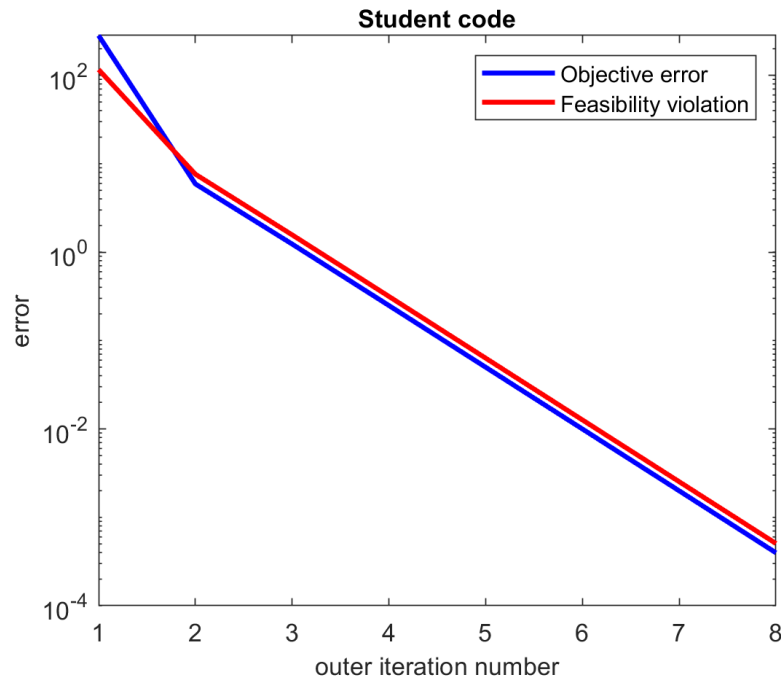


Figure 1: Student `test_penalty_qp` Result

Instructor Figure Result

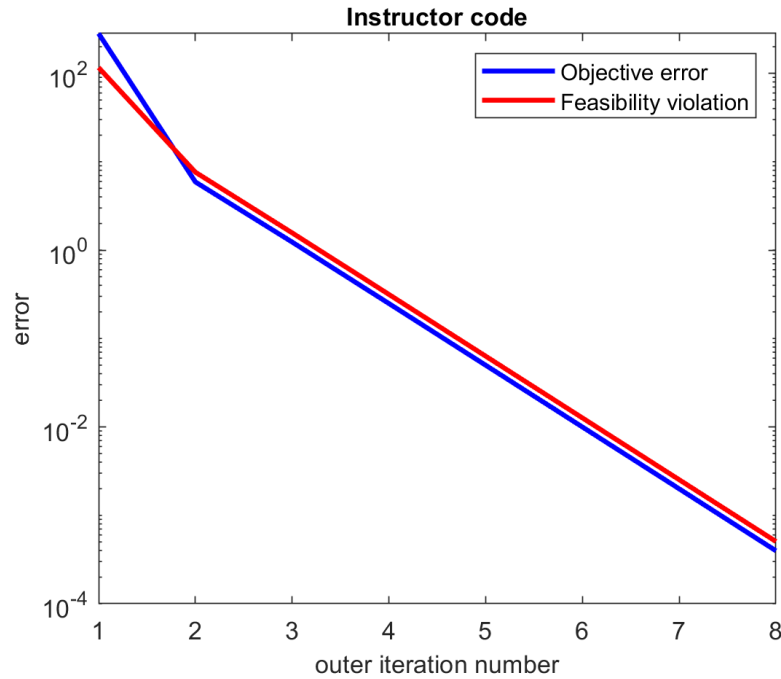


Figure 2: Instructor test_penalty_qp Result

Therefore, based on the plots of error vs iterations, we have nearly identical performances of functions, so we conclude our penalty_qp function is working correctly. Our student solver performs faster than the instructor, likely due to MatLab implementations, and our computation of AtA and Atb outside of the functions loops.

Problem 4

Consider the one-dimensional minimization problem

$$\min_x 2x - 1, \text{ s.t. } \frac{1}{2}x \leq 1, \quad x + 1 \geq 0 \quad (2)$$

1. Formulate the quadratic penalty problem with a penalty parameter $\mu > 0$ by penalizing **both** inequality constraints.

First, writing our problem in standard form

$$\begin{aligned} \min_x \quad & f(x) = 2x - 1 \\ \text{s.t.} \quad & g_1(x) = \frac{1}{2}x - 1 \leq 0 \end{aligned}$$

$$g_2(x) = -x - 1 \leq 0$$

Now we get our penalty problem,

$$\min_x f_\mu(x) = 2x - 1 + \frac{\mu}{2} \left(\left[\frac{1}{2}x - 1 \right]_+^2 + [-x - 1]_+^2 \right)$$

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2. Derive the optimal solution of the quadratic penalty problem. Note that the solution should depend on μ .

First, for finding the optimal solution we use:

$$\langle \nabla f_\mu(\bar{x}), x - \bar{x} \rangle \geq 0, \quad \forall x \in \mathbb{R}$$

$$\begin{aligned} \nabla f_\mu(x) &= 2 + \frac{\mu}{2} \left(\left[\frac{1}{2}x - 1 \right]_+ \right) + \frac{\mu}{2} (-2[-x - 1]_+) \\ &= 2 + \frac{\mu}{2} \left[\frac{1}{2}x - 1 \right]_+ + \mu [x + 1]_+ \end{aligned}$$

Now, we analyze this equation in each feasible region

Case 1 ($-1 \leq x \leq 2$) : (both penalties = 0)

$$f_\mu(x) = 2x - 1$$

$$\nabla f_\mu(x) = 2$$

Thus for case 1, $\nabla f_\mu(x) = 2 \implies$ no stationary point, so minimum occurs at $x = -1$

Case 2 ($x < -1$) : (Only g_2 is active):

$$f_\mu(x) = 2x - 1 + \frac{\mu}{2}(x + 1)^2$$

$$\nabla f_\mu(x) = 2 + \mu(x + 1)$$

$$\nabla f_\mu(x) = 0 \implies x = -\frac{2}{\mu} - 1$$

Thus for case 2, we have our minimum occur at $x = -\frac{2}{\mu} - 1$

Case 3 ($x > 2$): (only g_1 is active)

$$f_\mu(x) = 2x - 1 + \frac{\mu}{2} \left(\frac{x}{2} - 1 \right)^2$$

$$\nabla f_\mu(x) = 2 + \frac{\mu}{2} \left(\frac{x}{2} - 1 \right) = 2 + \frac{\mu}{4}(x - 2)$$

$$\nabla f_\mu(x) = 0 \implies x = -\frac{8}{\mu} + 2$$

Now, since $\mu > 0$, $x = -\frac{8}{\mu} + 2 < 2$. But, this case assumes $x > 2$. Thus, the minimum is not in this region.

Now, evaluating our potential options of $x = -1$ and $x = -\frac{2}{\mu} - 1$:

$$f(-1) = -3$$

$$f\left(-\frac{2}{\mu} - 1\right) = -3 - \frac{2}{\mu}$$

Since $\mu > 0$, $f\left(-\frac{2}{\mu} - 1\right) = -3 - \frac{2}{\mu} < -3$, so $f\left(-\frac{2}{\mu} - 1\right) < f(-1)$.

Thus, our optimal solution of the quadratic penalty problem is

$$\boxed{x_\mu = -\frac{2}{\mu} - 1}$$

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3. Prove that the solution of the quadratic penalty problem converges to the optimal solution $x^* = -1$ of Problem (2).

Proof. We will show

$$\lim_{\mu \rightarrow \infty} x_\mu = x^*, \quad \text{where } x^* = -1$$

From our derivation in part 2, the optimal solution to the penalty problem is

$$x_\mu = -1 - \frac{2}{\mu}$$

Taking the limit as the penalty parameter μ approaches infinity:

$$\lim_{\mu \rightarrow \infty} x_\mu = \lim_{\mu \rightarrow \infty} x_\mu = -1 - 0 = -1$$

So,

$$\lim_{\mu \rightarrow \infty} x_\mu = -1 = x^*$$

Thus, as $\mu \rightarrow \infty$ the quadratic penalty method arrives at the true constrained solution, $x^* = -1$.

□

Problem 5

Consider the problem

$$\min_{x,y \in \mathbb{R}} f(x,y) = -\frac{1}{2}x^2 - xy + \frac{1}{2}y^2 - 2x - 2y, \text{ s.t. } x + y = 1.$$

1. Argue that $(1,0)$ is the unique minimizer.

First, constraint $x + y = 1 \implies y = 1 - x$. Thus, subbing into $f(x,y)$:

$$\begin{aligned} f(x,y) &= f(x, 1-x) = -\frac{1}{2}x^2 - x(1-x) + \frac{1}{2}(1-x)^2 - 2x - 2(1-x) \\ &= -\frac{1}{2}x^2 - x + x^2 + \frac{1}{2}x^2 - x + \frac{1}{2} - 2x - 2 + 2x \\ &= x^2 - 2x - \frac{3}{2} \end{aligned}$$

Now we can find the minimum by setting the derivative equal to zero:

$$f'(x) = 2x - 2$$

$$f'(x) = 0 \implies 2x - 2 = 0 \implies x = 1$$

Now with $x = 1 \implies y = 1 - x = 1 - 1 = 0$

Therefore, $x = 1$ and $y = 0$, or $(1,0)$ is the unique minimizer of this problem.

2. The augmented Lagrangian function with penalty parameter $\beta > 0$ is

$$\mathcal{L}(x, y, v) = -\frac{1}{2}x^2 - xy + \frac{1}{2}y^2 - 2x - 2y + v(x + y - 1) + \frac{\beta}{2}(x + y - 1)^2.$$

Argue that for any $\beta \in (0, 1]$, the augmented Lagrangian method with initial multiplier $v^{(0)} = 0$ does not work for this instance, i.e., the generated sequence $(x^{(k)}, y^{(k)})$ does not converge to a KKT point.

First, we have

$$(x^{(k+1)}, y^{(k+1)}) = \arg \min_{(x, y)} \mathcal{L}_\beta(x, y, v^{(k)})$$

With

$$\mathcal{L}(x, y, v) = -\frac{1}{2}x^2 - xy + \frac{1}{2}y^2 - 2x - 2y + v(x + y - 1) + \frac{\beta}{2}(x + y - 1)^2$$

$$\nabla \mathcal{L}(x, y, v) = \begin{bmatrix} -x - y - 2 + v + \beta(x + y - 1) \\ -x + y - 2 + v + \beta(x + y - 1) \end{bmatrix}$$

$$\nabla^2 \mathcal{L}(x, y, v) = \begin{bmatrix} -1 + \beta & -1 + \beta \\ -1 + \beta & 1 + \beta \end{bmatrix}$$

$$\det(\nabla^2 \mathcal{L}) = (\beta - 1)(\beta + 1) - (\beta - 1)^2 = \beta^2 - 1 - \beta^2 + 2\beta - 1 = 2\beta - 2, \quad \text{with } \beta \in (0, 1]$$

Case 1: $\beta \in (0, 1)$

$$\det(\nabla^2 \mathcal{L}) = 2(\beta) - 2 < 2(1) - 2 = 0 \implies \det(\nabla^2 \mathcal{L}) < 0 \implies \nabla^2 \mathcal{L} \text{ is indefinite.}$$

Thus, $\mathcal{L}(x, y, v^{(k)})$ is non-convex so it is unbounded below, and

$$\min_{(x, y)} \mathcal{L}(x, y, v^{(k)}) = -\infty$$

So, for $\beta \in (0, 1)$ no minimizer exists and $x^{(k+1)}$ is not well-defined.

Case 2: $\beta = 1$

$$\det(\nabla^2 \mathcal{L}) = 2(1) - 2 = 0 \implies \text{Singular, not strictly convex}$$

Taking $\nabla \mathcal{L}$ at $v^{(0)} = 0$:

$$\begin{aligned} \nabla \mathcal{L}(x, y, 0) &= \begin{bmatrix} -x - y - 2 + 0 + \beta(x + y - 1) \\ -x + y - 2 + 0 + \beta(x + y - 1) \end{bmatrix} \\ &\implies \begin{bmatrix} -x - y - 2 + x + y - 1 \\ -x + y - 2 + x + y - 1 \end{bmatrix} \\ &\implies \begin{bmatrix} -3 \\ 2y - 3 \end{bmatrix} \end{aligned}$$

Now, $\frac{\partial \mathcal{L}}{\partial x} = -3 \neq 0$, so x continues to decrease and does not converge.

Thus for $\beta = 1$, $x \rightarrow -\infty$, so, $\mathcal{L}(x, y, v^{(k)}) \rightarrow -\infty$ and again no minimizer exists.

Therefore, for any $\beta \in (0, 1]$, $\min_{(x,y)} \mathcal{L}(x, y, v^{(k)})$ is unbounded below, so $x^{(k+1)}$ is not well defined and this method does not work.

There is no $\beta \in (0, 1]$ such that $\nabla_x \mathcal{L}(x^{(k+1)}, v^{(k)}) = 0$.

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3. For $\beta = 2$, do by hand two iterations of the augmented Lagrangian method with initial multiplier $v^{(0)} = 0$.

First, we have

$$\begin{cases} x^{(k+1)} = \arg \min_{x \in \mathbb{R}} \mathcal{L}(x, y, v^{(k)}) \\ y^{(k+1)} = \arg \min_{y \in \mathbb{R}} \mathcal{L}(x, y, v^{(k)}) \\ v^{(k+1)} = v^{(k)} + \beta(Ax^{(k+1)} - b) \end{cases}$$

Now, we get the argmins by setting the respective gradients to 0.

$$\begin{aligned} \nabla \mathcal{L}(x, y, v) &= \begin{bmatrix} -x - y - 2 + v + 2(x + y - 1) \\ -x + y - 2 + v + 2(x + y - 1) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \implies \begin{bmatrix} -x - y - 2 + v + 2x + 2y - 2 \\ -x + y - 2 + v + 2x + 2y - 2 \end{bmatrix} &= \begin{bmatrix} x + y - 4 + v \\ x + 3y - 4 + v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \implies 2y &= 0 \implies y^{(k+1)} = 0 \end{aligned}$$

$$\implies x + 3(0) - 4 + v = 0 \implies x^{(k+1)} = 4 - v^{(k)}$$

Now, $v^{(k+1)} = v^{(k)} + \beta(Ax^{(k+1)} - b)$

$$\implies v^{(k+1)} = v^{(k)} + 2(x^{(k+1)} + y^{(k+1)} - 1) = v^{(k)} + 2(x^{(k+1)} - 1)$$

Therefore, we have iterations defined as:

$$x^{(k+1)} = 4 - v^{(k)}$$

$$y^{(k+1)} = 0$$

$$v^{(k+1)} = v^{(k)} + 2(x^{(k+1)} - 1)$$

Iteration 1 ($k = 0$): (Starting with $v^{(0)} = 0$)

$$v^{(0)} = 0$$

$$x^{(1)} = 4 - 0 = 4$$

$$y^{(1)} = 0$$

$$v^{(1)} = 0 + 2(4 - 1) = 6$$

Iteration 2 ($k = 1$):

$$v^{(1)} = 6$$

$$x^{(2)} = 4 - 6 = -2$$

$$y^{(2)} = 0$$

$$v^{(2)} = 6 + 2(-2 - 1) = 0$$

Thus, after 2 iterations we have $\boxed{x^{(2)} = -2, y^{(2)} = 0}$, or $\boxed{(-2, 0)}$

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4. (**bonus question**) For any $\beta > 2$, does the generated sequence $(x^{(k)}, y^{(k)})$ converge to the unique minimizer $(1, 0)$, starting from any $v^{(0)}$? If yes, prove it; if not, give a value of $\beta > 2$ and $v^{(0)}$ such that the sequence does not converge to $(1, 0)$.

Yes. For any $\beta > 2$, the generated sequence $(x^{(k)}, y^{(k)})$ converges to the unique minimizer $(1, 0)$ for any initial $v^{(0)}$.

Proof. First,

$$\begin{cases} x^{(k+1)} = \arg \min_{x \in \mathbb{R}} \mathcal{L}(x, y, v^{(k)}) \\ y^{(k+1)} = \arg \min_{y \in \mathbb{R}} \mathcal{L}(x, y, v^{(k)}) \\ v^{(k+1)} = v^{(k)} + \beta(Ax^{(k+1)} - b) \end{cases}$$

We can get the argmins by setting the respective gradients to 0.

$$\begin{aligned} \nabla \mathcal{L}(x, y, v) &= \begin{bmatrix} -x - y - 2 + v + \beta(x + y - 1) \\ -x + y - 2 + v + \beta(x + y - 1) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \implies \begin{bmatrix} -x - y - 2 + v + \beta x + \beta y - \beta \\ -x + y - 2 + v + \beta x + \beta y - \beta \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \implies 2y = 0 &\implies y^{(k+1)} = 0 \end{aligned}$$

With $y^{(k+1)} = 0$

$$y^{(k+1)} = 0 \implies -x^{(k+1)} - 0 - 2 + v^{(k)} + \beta(x^{(k+1)} + 0 - 1) = 0$$

$$\implies x^{(k+1)}(\beta - 1) = \beta + 2 - v^{(k)}$$

$$\implies x^{(k+1)} = \frac{\beta + 2 - v^{(k)}}{\beta - 1}$$

With $x^{(k+1)} = \frac{\beta + 2 - v^{(k)}}{\beta - 1}$:

$$v^{(k+1)} = v^{(k)} + \beta(Ax^{(k+1)} - b)$$

$$= v^{(k)} + \beta(x^{(k+1)} + y^{(k+1)} - 1)$$

$$\begin{aligned}
&= v^{(k)} + \beta \left(\frac{\beta + 2 - v^{(k)}}{\beta - 1} - 1 \right) \\
&= v^{(k)} + \beta \left(\frac{\beta + 2 - v^{(k)} - (\beta - 1)}{\beta - 1} \right) \\
&= v^{(k)} + \beta \left(\frac{3 - v^{(k)}}{\beta - 1} \right) \\
&= \left(1 - \frac{\beta}{\beta - 1} \right) v^{(k)} + \frac{3\beta}{\beta - 1} \\
&= \left(-\frac{1}{\beta - 1} \right) v^{(k)} + \frac{3\beta}{\beta - 1}
\end{aligned}$$

Now, with $v^{(k+1)} = \left(-\frac{1}{\beta-1} \right) v^{(k)} + \frac{3\beta}{\beta-1}$, let v^* denote the limit of the sequence $\{v^{(k)}\}$. Taking limits on both sides of the recursion, we obtain:

$$\begin{aligned}
v^* &= \left(-\frac{1}{\beta - 1} \right) v^* + \frac{3\beta}{\beta - 1} \\
\implies v^* \left(1 + \frac{1}{\beta - 1} \right) &= \frac{3\beta}{\beta - 1} \\
\implies v^* \left(\frac{\beta}{\beta - 1} \right) &= \frac{3\beta}{\beta - 1} \implies v^* = 3
\end{aligned}$$

Thus, $v^{(k)} \rightarrow 3$. Now,

$$y^{(k+1)} = 0$$

$$x^{(k+1)} = \frac{\beta + 2 - y^{(k)}}{\beta - 1} \rightarrow \frac{\beta + 2 - 3}{\beta - 1} = \frac{\beta - 1}{\beta - 1} = 1$$

So, $x \rightarrow 1$.

Thus, for any $\beta > 2$,

$$v^{(k)} \rightarrow 3$$

$$y^{(k)} \rightarrow 0$$

$$x^{(k)} \rightarrow 1$$

So for any initial $v^{(k)}$, our sequence will always converge to $x^{(k)} = 1$, $y^{(k)} = 0$, or $(1, 0)$, the minimizer.

□